



MEMORANDUM

To: Professor Brendon Anderson, Mechanical Engineering, California Polytechnic State University, bga@calpoly.edu
From: Gabriel Choi, Carmen Alaichamy
Date: May 20, 2026
Subject: Lab 5A: Sine Sweep Test

Introduction

The objective of this lab was to experimentally determine the frequency response of the pendulum system using a sine sweep input and develop a second-order transfer function model from the measured Bode plot data.

Methodology

The experiment was conducted using an STM32 microcontroller connected to a DC motor and rotary encoder. A linear sine sweep input signal was applied to the pendulum system to excite a range of frequencies. The angular response of the pendulum was measured using the encoder and recorded in Jupyter Notebook. The recorded input and output data were used to generate experimental Bode plots for magnitude and phase. A second-order transfer function model was then tuned by adjusting ω_n , ζ , and K_{ss} until the theoretical response closely matched the experimental data.

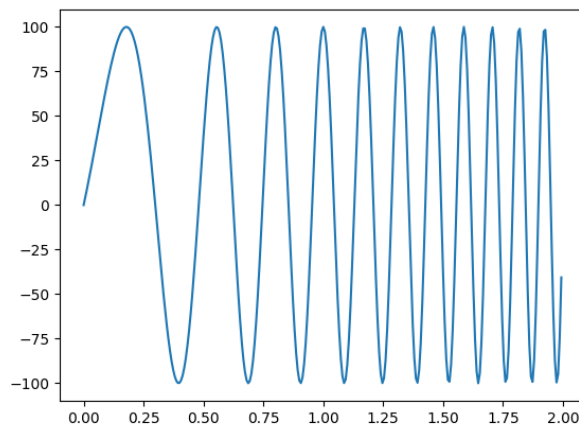


Figure 1. Linear Sine Sweep

Results

The following plots in Figure 2 shows the theoretical (red) and experimental (blue) Bode plots.

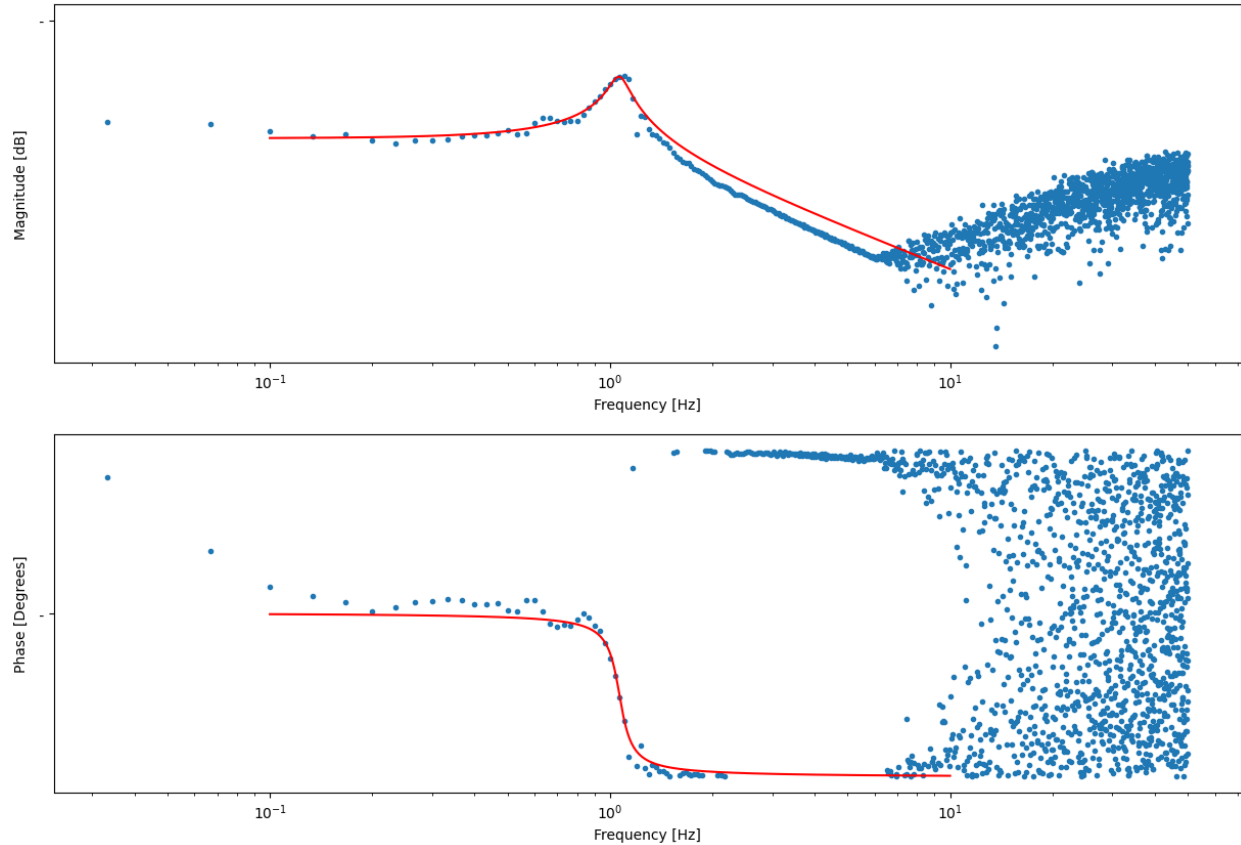


Figure 2. Experimental and Theoretical Overlaid Bode Plots

The experimental Bode plots showed a resonant peak near the natural frequency of the pendulum system. The second-order transfer function model provided a reasonable approximation of both the magnitude and phase response in the reliable frequency range. The resonance behavior and phase lag observed experimentally were captured by selecting appropriate values of ω_n , ζ , and K_{ss} , with units of radians per second, dimensionless units (-), and rad^{-1} respectively.

```
wn = 6.7;  
zeta = 0.41;  
Kss = 0.018;  
  
num = Kss*wn^2;  
den = [1 2*zeta*wn wn^2];  
  
G = tf(num, den);  
  
bode(G)
```

grid on

Figure 3. MATLAB Model Transfer Function to Generate the Theoretical Curve.

The equation below shows the final transfer function.

$$\frac{0.018 * 6.7^2}{s^2 + 2 * 0.41 * 6.7s + 6.7^2}$$

Discussion

Justify how ω_n , ζ , and K_{ss} affect the shape of the Bode plots.

K_{ss} affects the overall gain of the system and vertically shifts the magnitude plot. Increasing K_{ss} raises the magnitude response across all frequencies while decreasing K_{ss} lowers the response.

The damping ratio ζ controls the sharpness of the resonance peak. Lower damping produces a larger resonant peak and greater oscillatory behavior, while higher damping smooths the response and reduces overshoot.

The natural frequency ω_n determines the location of the resonance peak and shifts the frequency response horizontally along the frequency axis.

What frequency range can we consider reliable in the experiment? Hint: Check the frequency range of the sweep signal.

The frequency range that we consider reliable for this experiment is within the range of 2-6 hz for the long pendulum. This is because outside of that range we would encounter constructive interference which would cause our pendulum to oscillate outside of the range of $-\pi < \theta < \pi$.

Why does the data look very chaotic after 10Hz? How can we improve the frequency response after 10HZ?

After 10 Hz, the data becomes chaotic because the pendulum and motor cannot respond accurately to very fast input signals. At high frequencies, sensor noise, sampling limitations, and mechanical vibrations become more significant, making the measurements unreliable. The frequency response could be improved by increasing the sampling rate, using better filtering techniques, and improving the hardware or sensor accuracy.

Conclusion

The experiment successfully characterized the frequency response of the pendulum system using experimental Bode plots. A second-order transfer function model was developed that reasonably

matched the measured magnitude and phase responses. The lab demonstrated how ω_n , ζ , and K_{ss} affect system dynamics and frequency-domain behavior.